③ $a = 2$; $b = 8$; $N = 28$

$U = \{28, 29, 30, \ldots, 48\}$

$A = \{28, 30, 32, 35, 38, 41, 43\}$

$B = \{31, 33, 35, 38, 40, 46\}$

$C = \{29, 35, 38, 39, 45, 48\}$

$A \cup B \cup C$

$= \{28, 29, 30, 31, 32, 33, 35, 38, 39, 40, 41, 43, 45, 46, 48\}$

$A \cap B \cap C$

$= \{35, 38\}$

$C' = U - C$

$= \{28, 30, 31, 32, 33, 34, 36, 37, 40, 41, 42, 43, 44, 46, 47\}$

$(A \cup B) \cap C'$

$= \{28, 30, 31, 32, 33, 40, 41, 43, 46\}$

SiDU

Dipindai dengan CamScanner

$f(x) = Kx + (a+b) = 11x + 10$

$g(x) = \frac{x^2}{b+1} + a = \frac{x^2}{9} + 2$

$(f \circ g)(x) = 11\left(\frac{x^2}{9} + 2\right) + 10$

$= \frac{11x^2}{9} + 22 + 10 = \frac{11x^2}{9} + 32$

$(g \circ f)(x) = \frac{(11x+10)^2}{9} + 2$

$= \frac{121x^2 + 220x + 100}{9} + 2$

$= \frac{121x^2 + 220x + 100}{9} + \frac{18}{9} = \frac{121x^2 + 220x + 118}{9}$

tidak komutatif karena hasil beda

bukti numerik : $x = b+2$

$= 8+2 = 10$

$(f \circ g)(10) = \frac{1100}{9} + 32 = 122,22 + 32 = 154,22$

$(g \circ f)(10) = \frac{12100 + 2200 + 118}{9} = \frac{14418}{9} = 1602$

$y = 11x + 10$

$11x = y - 10$

$f^{-1}(y) = \frac{y-10}{11}$

Pendapatan total :

$K \cdot 100 + a + b = 1100 + 2 + 8 = 1110$

$x = f^{-1}(1110) = \frac{1110 - 10}{11} = \frac{1100}{11} = 100$

Pembelian = 100

	f(x)	g(x)
injektif	✓	×
surjektif	✓	×
bijektif	✓	×

Dipindai dengan CamScanner